

# Cloud Economy & Pricing

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A practical introduction to AWS pricing and cost estimation

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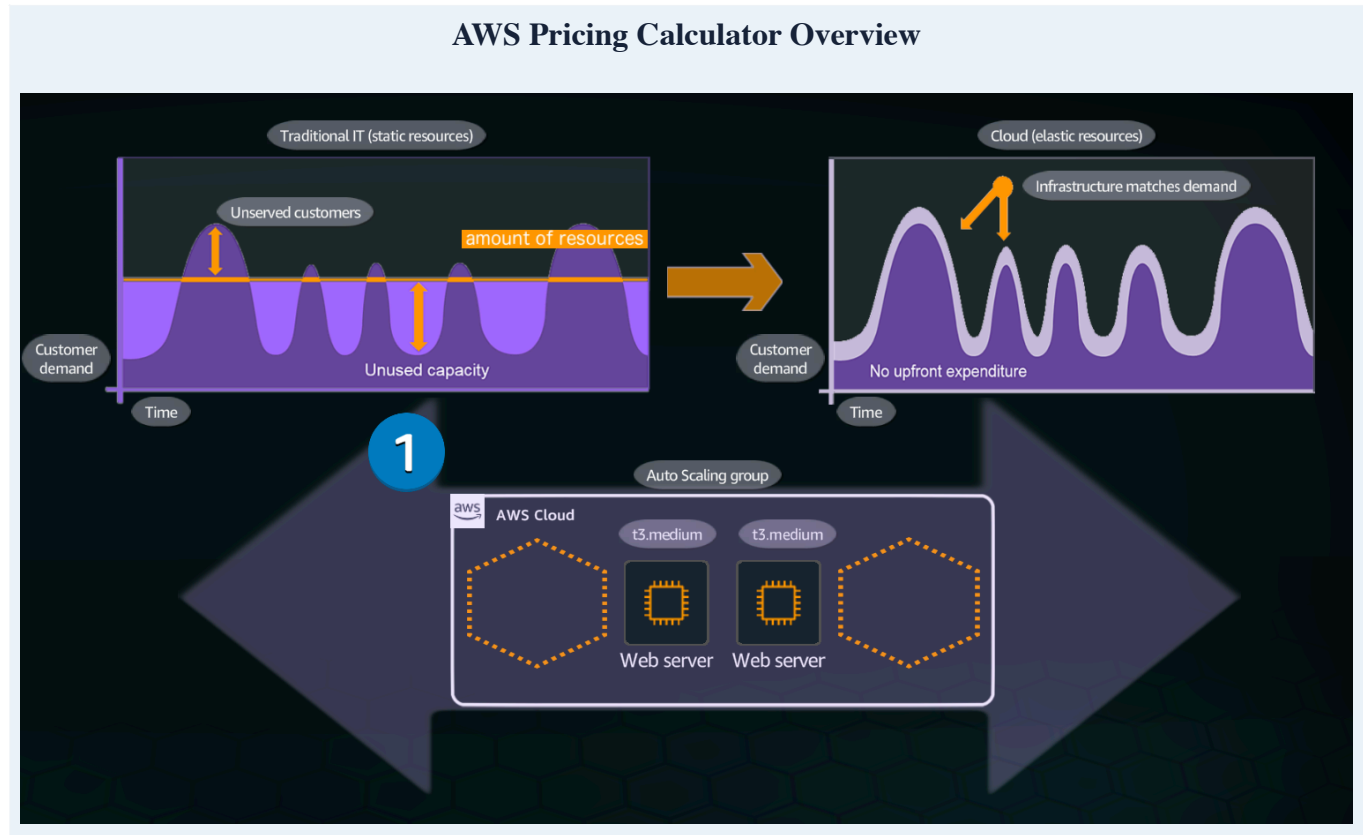
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Build and review an estimate with AWS Pricing Calculator.

# 01 Cloud Economy

Cloud economics is about matching infrastructure cost to actual use. Instead of buying enough physical hardware for every possible peak, AWS lets us consume resources when needed and estimate their cost before deployment.

## AWS Pricing Calculator

AWS Pricing Calculator creates estimates for AWS use cases. It can be used to model a solution, explore price points and review a detailed cost calculation before implementation.



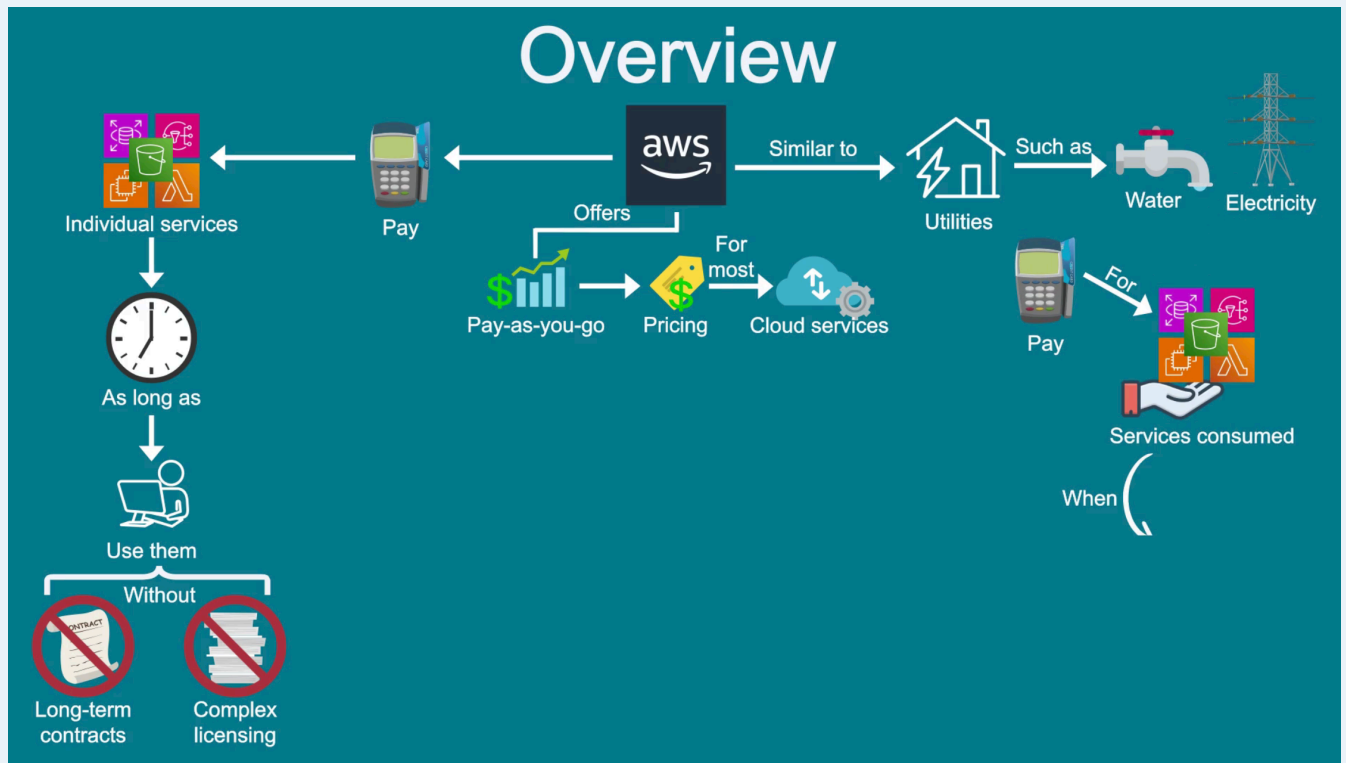
*Use the calculator before deployment to explore how architectural choices affect estimated cost.*

## AWS - Pay as you go

AWS uses a pay-as-you-go approach for most cloud services. You pay for the individual services you need, for as long as you use them, without having to buy the underlying infrastructure.

The idea is similar to utilities such as water or electricity: usage changes, so the bill changes with it.

## Pay-as-you-go overview



*Cloud spending follows resource consumption instead of requiring the full infrastructure cost up front.*

## Key principles

### *“Start early with cost optimisation”*

Cloud lets us trade fixed expenses, such as data centers and physical servers, for variable expenses based on what is consumed.

| Traditional                                    | Cloud                                   |
|------------------------------------------------|-----------------------------------------|
| Buy €50,000 of servers.                        | Use €200 of cloud resources this month. |
| The cost remains even when capacity sits idle. | Cost can follow actual resource usage.  |

#### **Fixed vs Variable - Cost**

Traditional infrastructure asks how much capacity we should buy. Cloud infrastructure asks how much capacity we need right now.

### *“Maximise the power of flexibility”*

AWS services are priced independently and are available on demand, so resources can be selected and adjusted around the needs of the workload.

A simple example is EC2.

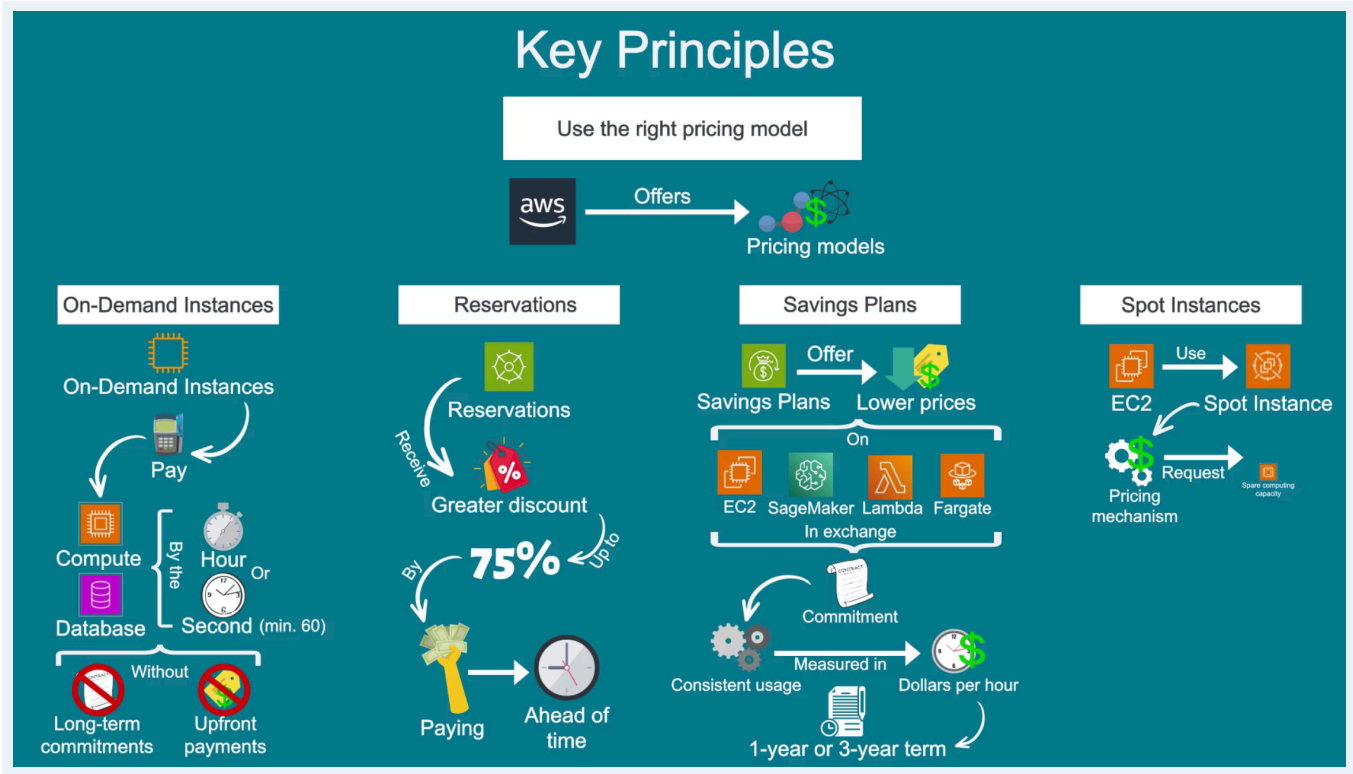
If an instance is not needed, stopping it stops the instance compute charge. Attached storage and some other resources can still continue to incur charges.

#### **Cost Control - unused resources matter**

Cloud makes it easier to reduce unused compute, but cost optimisation still requires monitoring what remains provisioned.

## Use the right pricing model

AWS offers different pricing models depending on the service. The right choice depends on how predictable the workload is and how much flexibility you need.



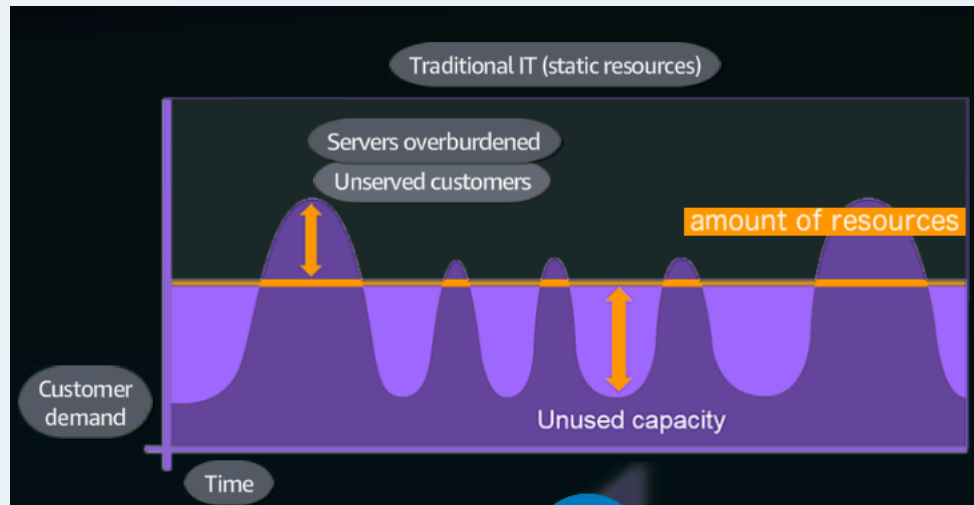
*Different models exchange flexibility, commitment and spare capacity for different price levels.*

| Pricing model | Beginner mental model                                                            |
|---------------|----------------------------------------------------------------------------------|
| On-Demand     | Use compute without a long-term commitment and pay for the usage.                |
| Reservations  | Commit to eligible capacity or usage to receive a lower effective price.         |
| Savings Plans | Commit to a consistent amount of eligible usage for a 1- or 3-year term.         |
| Spot          | Use spare EC2 capacity at a large discount, but the capacity can be interrupted. |

## Demand changes over time

Using fixed resources while demand changes creates two opposite problems. During high demand, the available capacity can become overloaded. During low demand, unused capacity creates unnecessary cost.

### Demand fluctuation



*Fixed capacity can be too small during peaks and too large during quiet periods.*

## Elasticity - Scaling capacity with demand

AWS resources can be adjusted as demand changes. During increased demand, a system can scale out horizontally by adding resources such as EC2 instances. When demand falls, active capacity can be reduced again.

`[EC2, EC2] → [EC2, EC2, EC2, EC2] → [EC2, EC2]`

EC2 Auto Scaling can automate this adjustment, helping infrastructure capacity follow workload demand more closely.

## What to remember

- Pay-as-you-go shifts infrastructure spending toward usage-based cost.
- Flexibility lets resources change instead of remaining permanently sized for peak demand.
- Pricing models trade flexibility and commitment for different levels of savings.
- Scaling helps align capacity with demand.
- AWS Pricing Calculator estimates an architecture before it is deployed.

**Use what you need, scale with demand, choose the pricing model that fits the workload, and estimate the cost before building.**

## AWS Pricing Calculator

The calculator does not create AWS resources or commit you to a purchase. It gives you a planning estimate so you can compare configurations, explore cost-saving opportunities and understand which parts of an architecture contribute to the expected bill.

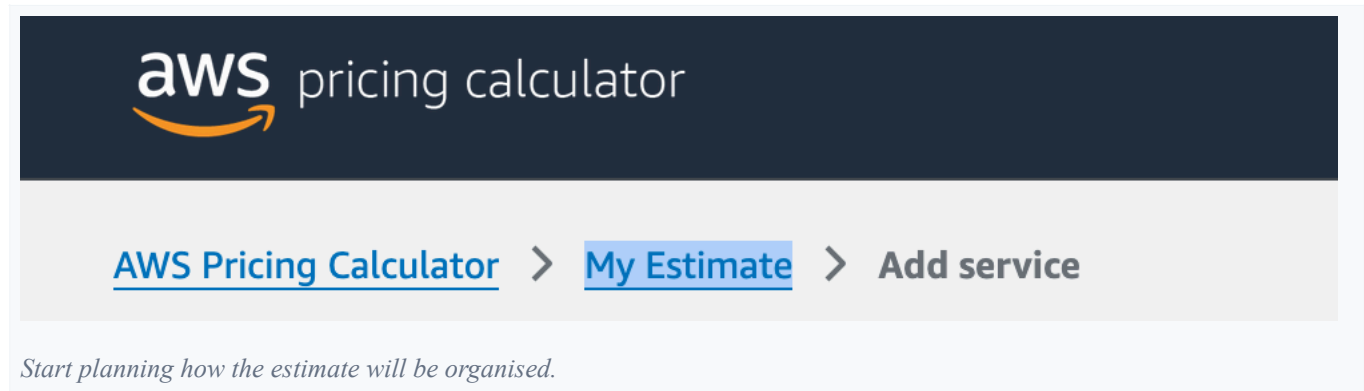
## 02 Practice Lab

### *Building an AWS cost estimate*

The theory above explains how AWS pricing works. In this practical lab, we use AWS Pricing Calculator to build an EC2 web-server estimate and see how workload, resource size, pricing model, storage and data transfer affect the expected cost.

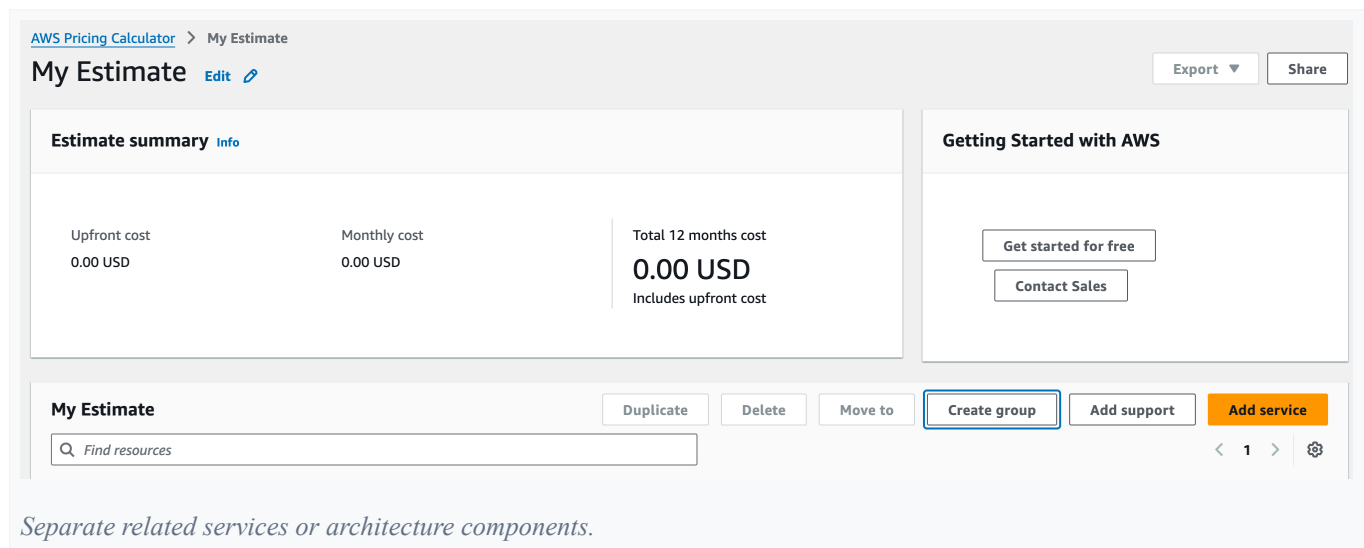
Start the estimate

Navigate to AWS Pricing Calculator and create an estimate. The My Estimate view helps organise estimates by cost centre, product stack or solution architecture.



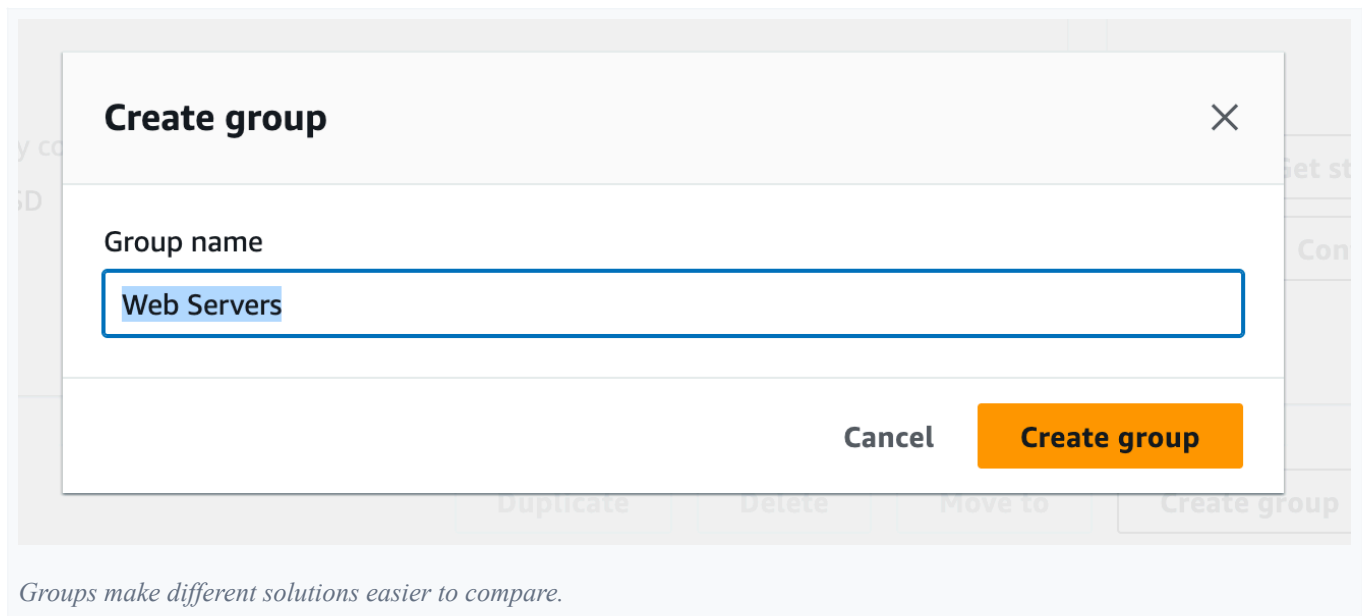
### Create groups

You can create different groups to compare cost estimates for different AWS setups.



*Separate related services or architecture components.*

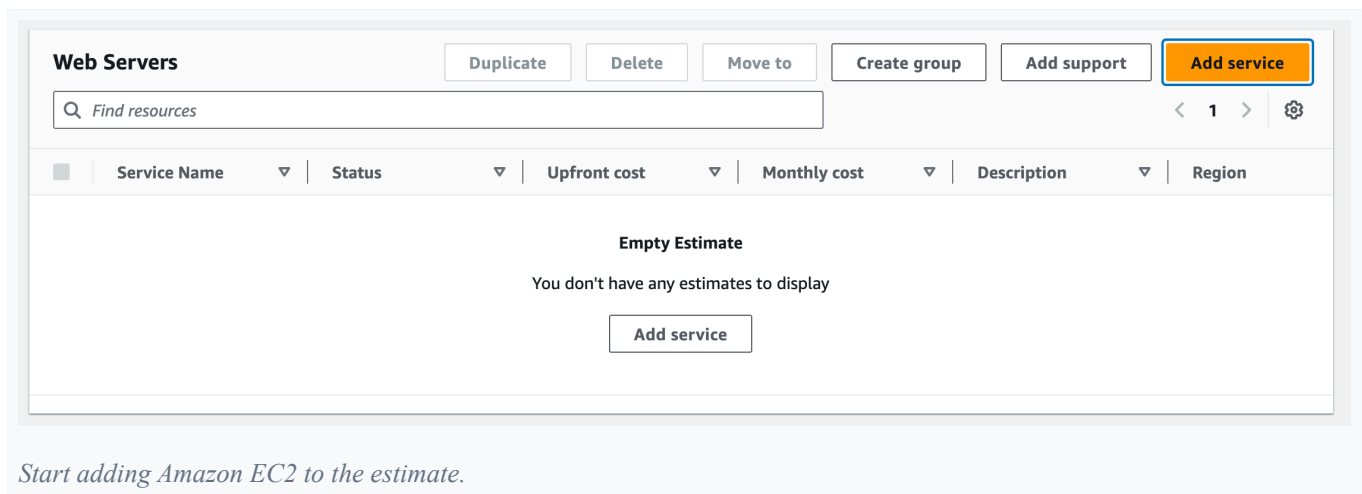




### Add Amazon EC2

AWS Pricing Calculator provides cost estimates for more than 175 AWS services and support options. Add a new EC2 service, then choose US East (N. Virginia).

Service costs vary by AWS Region, so select the Region carefully for an accurate estimate. In this example, we estimate a web server.



## Find EC2

**Add service** [Info](#)

**Bulk import**

**AWS services (11)** Cancel

☒ **Search by location type**  
See the services that are available in your region, wave length zone, and local zone.

☐ **Search all services**  
Choose a service or workload to configure an estimate.

Choose a location type [Info](#)

Choose a Region

Region

US East (Ohio)

Find Service

*Search for Amazon EC2.*

## Select EC2

### Amazon EC2

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instance types comprise varying combinations of CPU, memory, storage, and networking capacity and give you the flexibility to choose the appropriate mix of resources for your applications.

[Product page](#)

**Configure**

*Open the EC2 configuration.*

## Choose a Region

### Create estimate: Configure Amazon EC2 [Info](#)

Description

Web Server Estimate

Choose a location type [Info](#)

Region

Choose a Region

US East (N. Virginia)

*This example uses US East (N. Virginia).*

## Tenancy

Our first choice is tenancy: how the underlying physical hardware is allocated.

| Tenancy            | Meaning                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------|
| Shared / Default   | Your EC2 runs on physical hardware that may also host isolated EC2 instances from other customers. |
| Dedicated Instance | The physical hardware is dedicated to instances from your AWS account.                             |
| Dedicated Host     | You reserve an entire physical server with greater control over how instances use it.              |

AWS can split one powerful physical server into many isolated virtual servers. Shared tenancy lets multiple customers use the same hardware without sharing access to each other's instances. For most workloads, Shared / Default is the standard and cheaper choice.

## Choose tenancy

### EC2 specifications [Info](#)

#### Tenancy

Choose the tenancy type to run your Amazon EC2 instances on.

Shared Instances

*Tenancy affects how the underlying hardware is shared and can change the price.*

## Workloads

A workload describes how EC2 usage changes over time. Do you need the same amount of capacity all the time, or does demand increase at certain times?

| Workload              | What it means                                | Example                           |
|-----------------------|----------------------------------------------|-----------------------------------|
| Constant usage        | Roughly the same capacity all month          | Backend server running 24/7       |
| Daily spike traffic   | Usage increases at certain times every day   | Website busy during daytime       |
| Weekly spike traffic  | Usage increases on particular days each week | Business software busiest Mon-Fri |
| Monthly spike traffic | Usage increases at certain times each month  | Payroll system busy at month-end  |

### Choose a workload

#### Workloads

Choose the graph that best represents your monthly workload

☒ Constant usage ☐ Daily spike traffic ☐ Weekly spike traffic ☐ Monthly spike traffic

*Model how EC2 usage changes over time.*

### Workload example: an online store

An online store may be quiet overnight but much busier during the day and evening. That creates a daily spike because the busy hours repeat each day.

- Constant usage: relatively steady traffic throughout the month.
- Daily spike: busier during certain hours each day.
- Weekly spike: busiest on particular days, such as weekends.
- Monthly spike: activity increases around payday or month-end.

The important part is that the application type does not determine the workload pattern. How and when people use it does.

### Daily spike traffic

#### Workloads

Choose the graph that best represents your monthly workload

☐ Constant usage ☒ Daily spike traffic ☐ Weekly spike traffic ☐ Monthly spike traffic

▼ Daily spike pattern

Remove pattern

#### Workload days

Select days for your workload pattern.

- ☐ Sunday  
☒ Monday  
☒ Tuesday  
☒ Wednesday  
☒ Thursday  
☒ Friday  
☐ Saturday

*Model recurring busy hours for the online store.*

## Baseline and peak

- Baseline: the minimum number of EC2 instances normally needed.
- Peak: the maximum number needed when demand increases.
- Duration of peak: how long the workload stays at that maximum.

**Example:** the store normally runs 2 instances, scales up to 4 during busy hours, and stays at peak for 8 hours and 30 minutes.

### Baseline and peak instances

#### Baseline

Enter the minimum number of instances for your workload.

#### Peak

Enter the maximum number of instances for your workload.

#### Duration of peak (hours, minutes)

Enter the amount of days, hours, and minutes your instances are running at peak.

Hours

Minutes

*Show the always-on baseline and the extra instances used during peak demand.*

## Estimate the resources

We also need to estimate how much compute the workload might use.

| Resource     | What it affects                                             | How do I know what I need?                                                   |
|--------------|-------------------------------------------------------------|------------------------------------------------------------------------------|
| vCPU         | How much processing work the instance can handle            | More users, calculations, requests or CPU-heavy work may require more vCPUs. |
| Memory (GiB) | How much data and app state can stay in fast working memory | Large apps, caching or many processes may require more memory.               |

### A simple rule when starting

Start small and measure. CPU constantly high? Consider more vCPUs. Running out of memory? Consider more RAM. If everything runs comfortably, you probably do not need a larger instance.

You generally do not know the perfect EC2 size beforehand. Make an educated estimate, monitor the workload, and right-size it later.

How do you want to pay for EC2?

| Pricing option             | Simple explanation                                                                        | Good for                                                |
|----------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------|
| On-Demand                  | Pay for what you use with no long-term commitment                                         | Uncertain or changing workloads, learning               |
| Compute Savings Plans      | Commit to a certain amount of compute usage for 1 or 3 years in exchange for lower prices | Long-running workloads where you still want flexibility |
| EC2 Instance Savings Plans | Commit more closely to an EC2 instance family in a Region                                 | Predictable EC2 workloads                               |
| Spot Instances             | Use AWS spare EC2 capacity at a large discount, but AWS can interrupt it                  | Work that can tolerate interruptions                    |

## Choose a pricing option

☒ **Compute Savings Plans**

One plan that automatically applies to all usage on EC2, Fargate, and Lambda. Up to 66% discount. [Learn more](#)

Reservation term

☐ 1 year

☒ 3 year

Payment Options

☒ No upfront

☐ Partial upfront

☐ All upfront

☐ **EC2 Instance Savings Plans**

Get deeper discount when you only need one instance family and region. Up to 72% discount. [Learn more](#)

Reservation term

☐ 1 year

☒ 3 year

Payment Options

☒ No upfront

☐ Partial upfront

☐ All upfront

☐ **On-Demand**

Maximize flexibility. [Learn more](#)

☐ **Spot Instances**

Minimize cost by leveraging EC2's spare capacity. Recommended for fault tolerant and interruption tolerant applications. [Learn more](#)

The historical average discount for t3.medium is 60%

Assume percentage discount for my estimate

**i Actual spot instance pricing varies**

With spot instances, you pay the spot price that's in effect for the time period your instance is running

*Compare the EC2 pricing models available for the estimate.*

## The easiest way to understand them

Imagine our online store.

- On-Demand: the store is new, so we pay for EC2 as we use it without a long-term commitment.
- Compute Savings Plans: usage is predictable, so we commit to consistent compute spend for 1 or 3 years in return for a discount.
- EC2 Instance Savings Plans: we are more certain about the instance family and Region, giving up some flexibility for potentially greater savings.
- Spot Instances: use spare AWS capacity cheaply, but only for workloads that can tolerate interruption.

## Useful mental model

More commitment -> less flexibility -> potentially greater savings. Spot is different: the discount comes from using spare capacity AWS can take back.

No upfront, Partial upfront and All upfront describe when the commitment is paid. Paying more upfront can result in a lower effective price.

Reserved Instances can also provide billing discounts when you commit to 1-year or 3-year terms based on selected instance specifications.

Review workload hours

You can review workload hours per day and total instance utilisation per month. In this example, instances 1-2 run 24 hours per day while instances 3-4 are used only during the daily peak.

## Navigate to workload hours

### ▼ Show calculations

#### Utilization summary

View a complete calculation of your [estimated workload hours](#).

On-Demand instance hours: 1946.6666

1946.6666 On-Demand instance hours x 0.0416 USD = 80.981331 USD

**On-Demand instances (monthly): 80.981331 USD**

*Open the estimated EC2 utilisation.*

## Hours per day

### Workload hours per day

This table calculates the total instance utilization hours per day for your estimated daily workload.

|                         | Sunday | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|-------------------------|--------|--------|---------|-----------|----------|--------|----------|
| Baseline (count)        | 2      | 2      | 2       | 2         | 2        | 2      | 2        |
| Peak (count)            | 4      | 4      | 4       | 4         | 4        | 4      | 4        |
| Minutes                 | 0      | 0      | 0       | 0         | 0        | 0      | 0        |
| Hours                   | 8      | 8      | 8       | 8         | 8        | 8      | 8        |
| Total duration in hours | 8      | 8      | 8       | 8         | 8        | 8      | 8        |
| Baseline+Peak           | 4      | 4      | 4       | 4         | 4        | 4      | 4        |

*See when baseline and peak instances are active.*

## Monthly utilisation

### Workload hours per month

This table calculates the total instance utilization hours per month for your estimated daily workload.

< 1 ... >

| Instance c... | Sunday h... | Monday ... | Tuesday ... | Wednesd... | Thursday ... | Friday Ho... | Saturday ... | Total hours pe... | Total hours per ... |
|---------------|-------------|------------|-------------|------------|--------------|--------------|--------------|-------------------|---------------------|
| 1             | 24          | 24         | 24          | 24         | 24           | 24           | 24           | 168               | 730                 |
| 2             | 24          | 24         | 24          | 24         | 24           | 24           | 24           | 168               | 730                 |
| 3             | 8           | 8          | 8           | 8          | 8            | 8            | 8            | 56                | 243.3333            |
| 4             | 8           | 8          | 8           | 8          | 8            | 8            | 8            | 56                | 243.3333            |

*Compare always-on baseline instances with temporary peak instances.*

## Amazon Elastic Block Storage

EBS is persistent storage for EC2 instances. Think of it like a virtual SSD or hard drive attached to your EC2 server.

General Purpose SSD works well for basic web servers, while workloads with heavier input-output or data-transfer needs may require different storage performance.

A normal web server may occasionally read files, write logs or load application data. A busy database may perform thousands of small reads and writes, making storage performance the bottleneck even when CPU and RAM are sufficient.

| Need                                   | Why different storage might help      |
|----------------------------------------|---------------------------------------|
| Normal web server                      | General Purpose SSD is usually enough |
| Busy database                          | May need very high IOPS               |
| Large sequential data                  | May need high throughput              |
| Cheap, infrequently accessed workloads | May prioritise cost over speed        |

**IOPS** = how many storage operations can happen per second.

**Throughput** = how much data can move per second.

## EBS performance

### Storage for each EC2 instance

Choose EBS volume storage type.

General Purpose SSD (gp3)



### General Purpose SSD (gp3) - IOPS

gp3 supports a max of 80,000 IOPS per volume

30

*Storage performance matters differently for small operations and large data transfers.*

## Storage amount and throughput

When configuring an EBS volume, consider both how much data you need to store and how quickly that data



needs to move. For a normal web server, reasonable defaults are usually enough to start; monitoring can tell you when to adjust later.

| Setting             | What it controls                           | How do I know what I need?                                                                           |
|---------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------|
| Throughput (MBps)   | How quickly data moves between EC2 and EBS | Start with the gp3 default for normal applications. Increase it for sustained large reads or writes. |
| Storage amount (GB) | How much data the EBS volume can store     | Estimate the OS, application, logs and other files, with room for growth.                            |

### EBS storage amount

General Purpose SSD (gp3) - Throughput  
gp3 supports a max of 2000 MBps per volume

Value

Unit

MBps

Storage amount

10

Unit

GB

Estimate both storage capacity and performance.

### Snapshot frequency

A snapshot is a point-in-time backup of an EBS volume. Snapshot frequency means how often you create that backup.

### Snapshot frequency

Snapshot Frequency

Weekly

Amount changed per snapshot

1

Unit

GB

Backup frequency is another cost input for the estimate.

## Data transfer

AWS resources often need to send data to each other and to users. Some transfers are free, while others cost money depending on where the data is located and how it travels.

**Simple rule:** the farther data has to travel across AWS infrastructure or out to the internet, the more important it becomes to check data-transfer pricing.

| Direction | Example                     |
|-----------|-----------------------------|
| Inbound   | Browser requests /products  |
| Inbound   | User submits a login form   |
| Inbound   | User uploads a photo        |
| Outbound  | Server returns product data |
| Outbound  | Server sends HTML or JSON   |
| Outbound  | User downloads a file       |

## Inbound data

Inbound data is data travelling from the customer to your application, such as requests, forms or uploads.

| Setting            | What it means                                            |
|--------------------|----------------------------------------------------------|
| Data transfer from | Where the incoming data originates, such as the internet |
| Enter amount       | How much incoming data you expect                        |
| Data amount        | The unit used to measure it, such as GB per month        |

## Inbound data transfer

### Inbound Data Transfer

Enter the data you expect to transfer into US East (N. Virginia)

Data transfer from

Enter Amount

Data amount

**Add inbound data transfer**

*Estimate where incoming data originates and how much enters AWS.*

## Outbound data

Outbound data is data sent from your AWS environment to another destination. For a web app, this includes HTML, API responses, images and files returned to users.

Data transferred out to the internet can incur additional costs because AWS carries that data from its infrastructure onto external networks.

## Outbound data transfer

### Outbound Data Transfer

Enter the data you expect to transfer out of US East (N. Virginia)

Data transfer to

Internet (0.05 USD ... ▼

Enter Amount

100

Data amount

GB per month ▼

**Add outbound data transfer**

*Estimate how much data the application sends from AWS to users or other destinations.*

### The lab's final mental model

The estimate is built from the same questions as the architecture: where does it run, how much capacity is needed, how long is it used, how is it priced, how much storage does it need, and how much data moves?

AWS Pricing Calculator turns those assumptions into an estimate before the resources are deployed.

[AWS Pricing Calculator](#) > My Estimate

**My Estimate** [Edit](#) [↗](#)

Export ▼

Share

You can finalise and share the estimates as a link.